SUPPLY CHAIN MANAGEMENT DASHBOARD

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Role:- Business Analyst Project

*Project Overview :-

- Developed a Supply Chain Management Dashboard to analyze operational efficiency, delivery performance, and inventory status.
- The project focuses on understanding cost distribution, supplier performance, transportation efficiency, and inventory levels.
- Multiple visualizations such as pie charts, line charts, scatter plots, and bar charts were used to present insights.
- Key Performance Indicators (KPIs) like Total Inventory Value, On-Time Delivery Rate, and Average Defect Rate help measure supply chain health.
- The dashboard enables data-driven decision making for improving logistics efficiency and reducing operational costs.

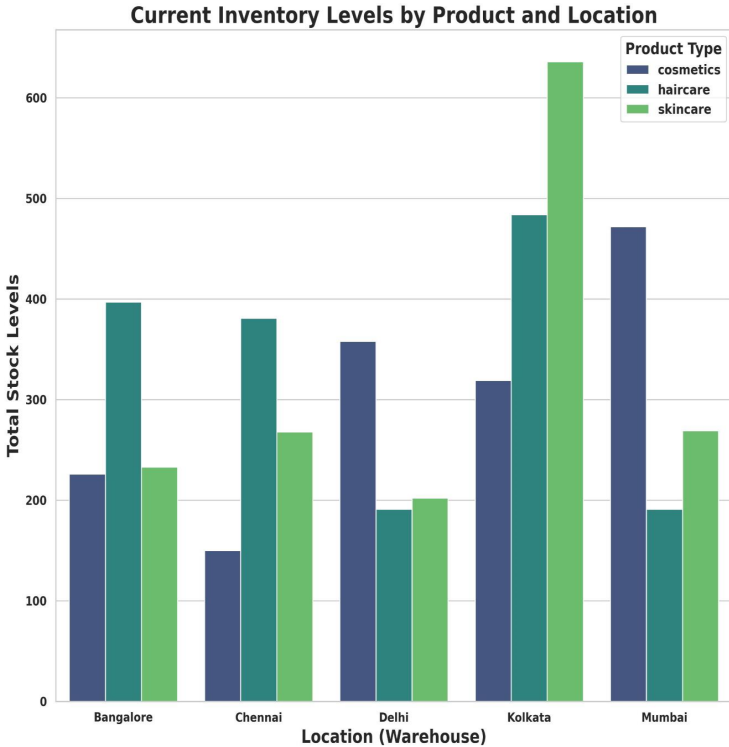
*Dataset Overview :-

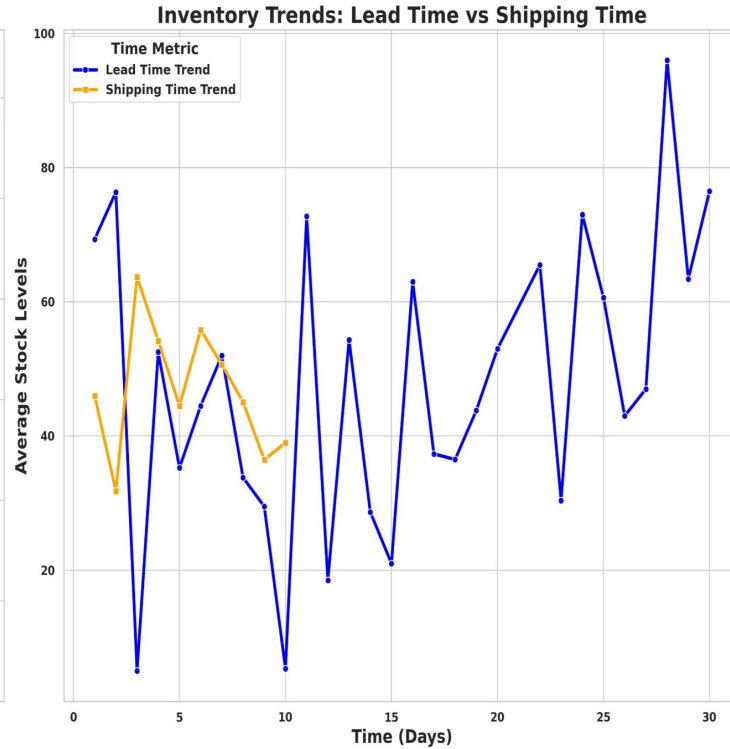
- The dataset contains information related to suppliers, shipping carriers, delivery status, inventory levels, and supply chain costs.
- It includes attributes such as lead time, shipping time, defect rate, product type, and warehouse location.
- Data records represent daily supply chain activities over a period of 30 days.
- The dataset also captures order fulfillment status (Pass, Fail, Pending) and on-time delivery rates for different product categories.
- It helps analyze inventory distribution across warehouses and performance metrics of suppliers and transportation carriers.

*KPI Cards :-

KPI'S	Values
Total Inventory Value	$5,000,000
On Time Delivery – Rate	95 %
Average Supplier Delivery Time	3 Days
Total Supply Chain Costs	$2,500,000
Average Defect Rate	2.28 %

Inventory Management Sheet

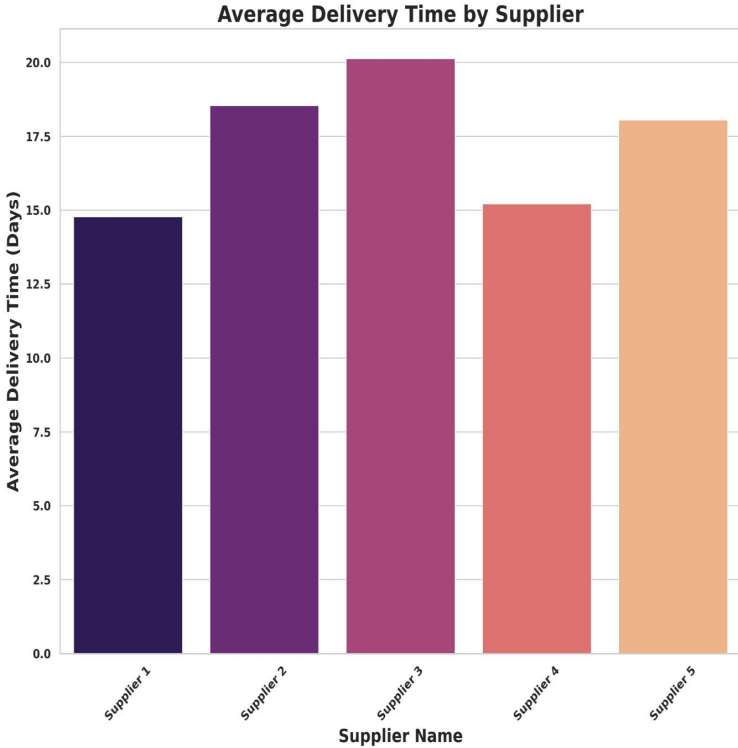


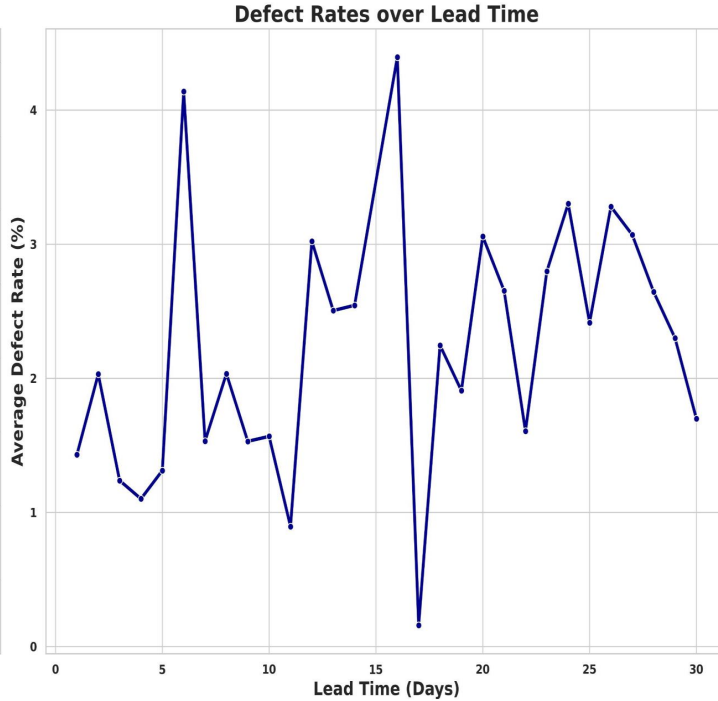


*Insights from Inventory Management Sheet :-

- Regional Stock Imbalance: Kolkata is significantly overstocked in Skincare compared to all other regions, while Chennai shows the lowest overall stock levels, suggesting a need for inventory redistribution.
- Product Category Dominance: Skincare is the primary volume driver in Kolkata and Delhi, whereas Cosmetics dominates the Mumbai warehouse, indicating distinct regional consumer preferences.
- Volatility in Lead Times: The “Lead Time Trend” shows extreme fluctuations (ranging from near 5 to 95), which suggests a highly unstable supply chain that could lead to frequent stockouts or overstocking.
- Short-Term Shipping Stability: Shipping times (orange line) appear much more stable and lower than lead times, suggesting that the primary bottleneck is in procurement or manufacturing rather than the final delivery leg.
- Stock-to-Time Correlation: There is a visible spike in lead times toward the 30-day mark; if this aligns with high-demand periods, the business faces a significant risk of being unable to fulfill orders.

Supplier Performance Sheet

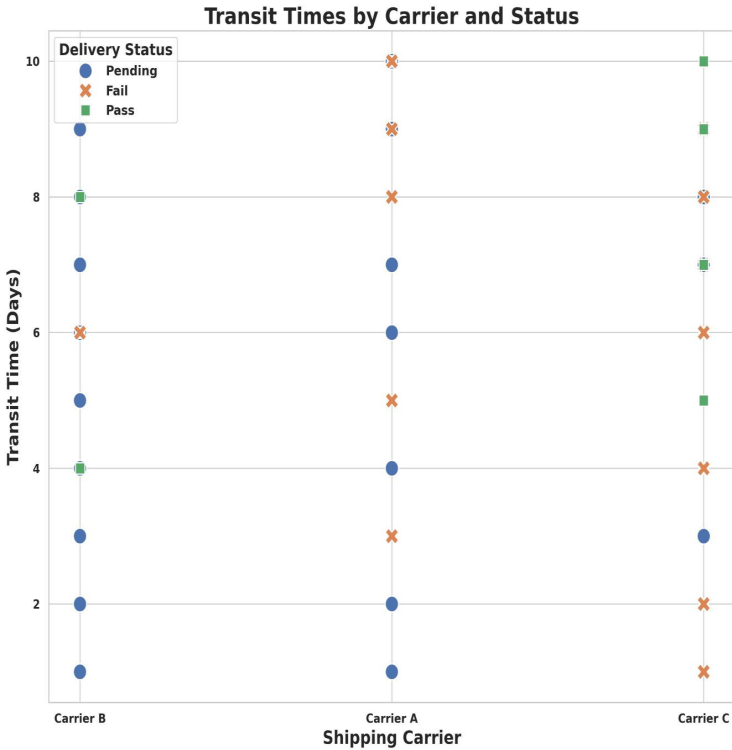


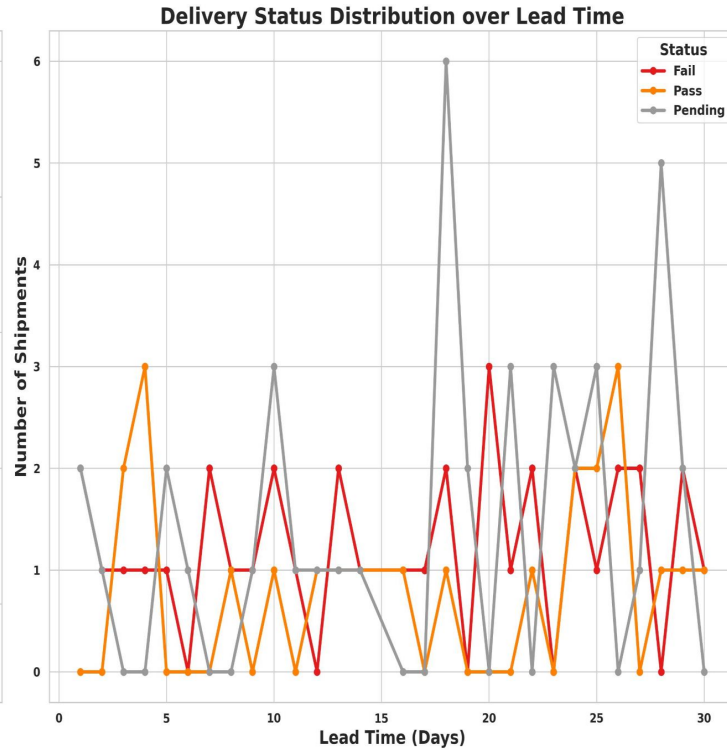


*Insights from Supplier Performance Sheet :-

- Supplier Efficiency Gap: Supplier 1 is your most efficient partner with the lowest average delivery time (~14 days), while Supplier 3 is the least efficient, exceeding 20 days.
- Critical Defect Spikes: There is a sharp increase in defect rates (peaking over 4%) when lead times hit approximately 6 and 16 days, indicating possible quality control breakdowns during "rush" or specific batch cycles.
- The "Sweet Spot" for Quality: The lowest defect rate (near 0%) occurs at a 17-day lead time. This suggests that certain lead times allow for optimal quality checks without being excessively long.
- Delivery vs. Quality Trade-off: While Supplier 1 is fast, the "Defect Rates over Lead Time" chart suggests that shorter lead times don't always correlate with higher defects, meaning speed doesn't have to sacrifice quality.
- Reliability Risk: The high variance in defect rates (0.2% to 4.4%) across different lead times suggests inconsistent manufacturing processes that require standardized quality agreements with all five suppliers.

Transportation Efficiency Sheet

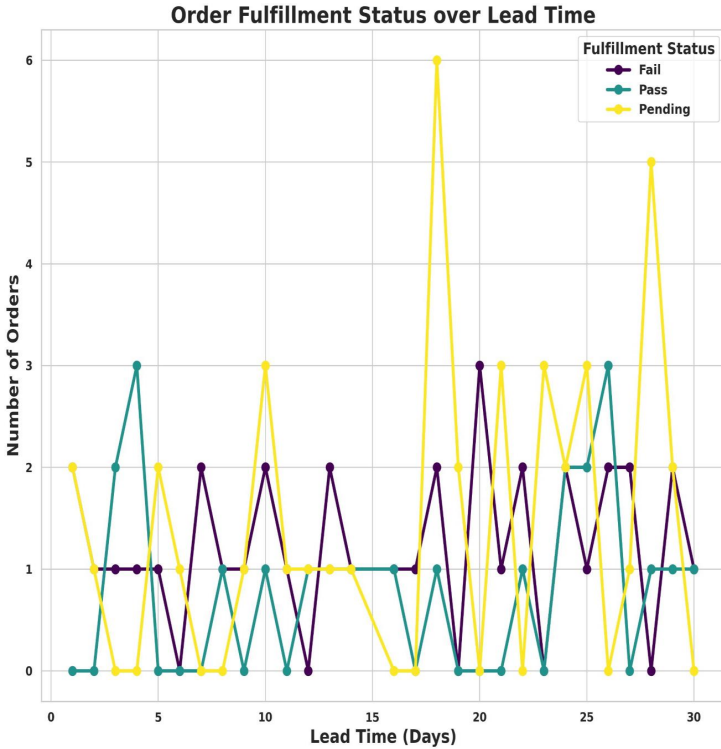


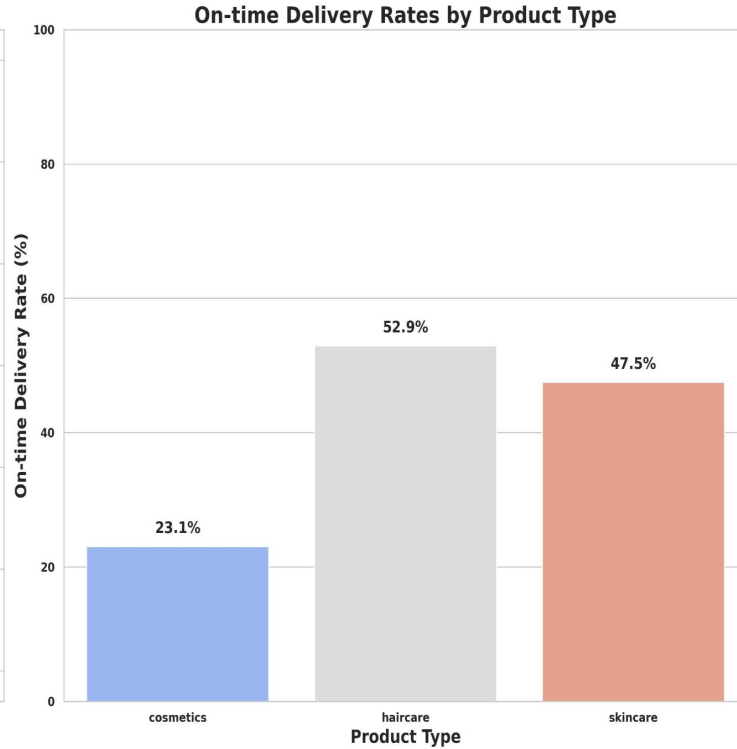


*Insights from Transportation Efficiency Sheet :-

- Carrier Reliability Gap: Carrier B shows a much higher concentration of “Pass” (green) statuses at lower transit times (4–10 days) compared to Carrier A, which has a heavy density of “Fail” (orange) results even at higher transit times.
- Transit Time Paradox: Surprisingly, longer transit times do not guarantee a “Pass” status. For Carrier C, failures occur at both the 1-day and 8-day marks, suggesting that the issue is likely operational handling rather than just speed.
- Pending Shipment Backlog: There is a massive spike in “Pending” shipments (gray line) around the 18-day and 28-day lead time marks. This indicates a recurring bottleneck in the middle and end of the month where logistics capacity likely hits its limit.
- Lead Time Failure Correlation: As lead time increases beyond 20 days, the frequency of “Fail” statuses (red line) becomes more volatile and frequent, suggesting that the longer a shipment stays in the system, the higher the risk of a delivery failure.
- Optimization Opportunity: Carrier B is the clear winner for speed and success. A strategic shift of high-priority volume from Carrier A to Carrier B could immediately improve regional delivery KPIs.

Order Fulfillment Sheet

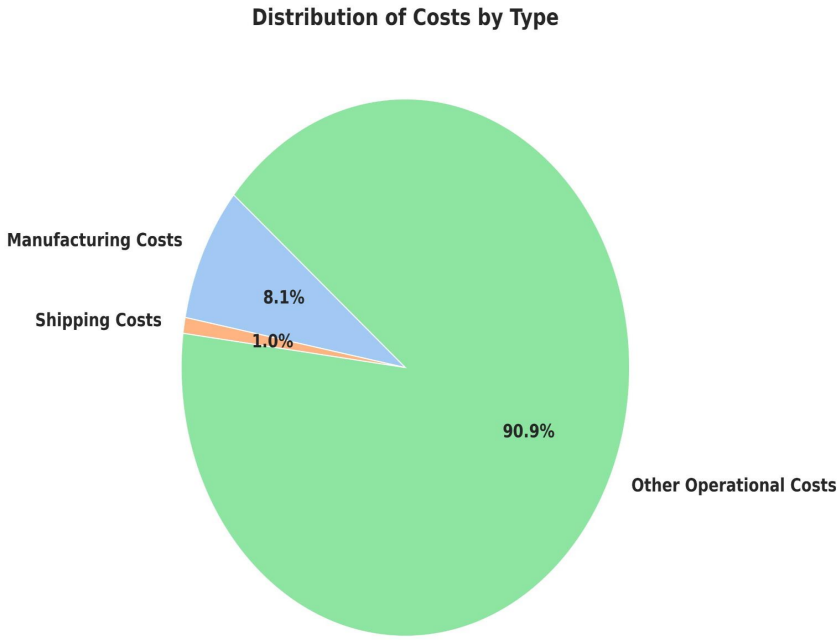


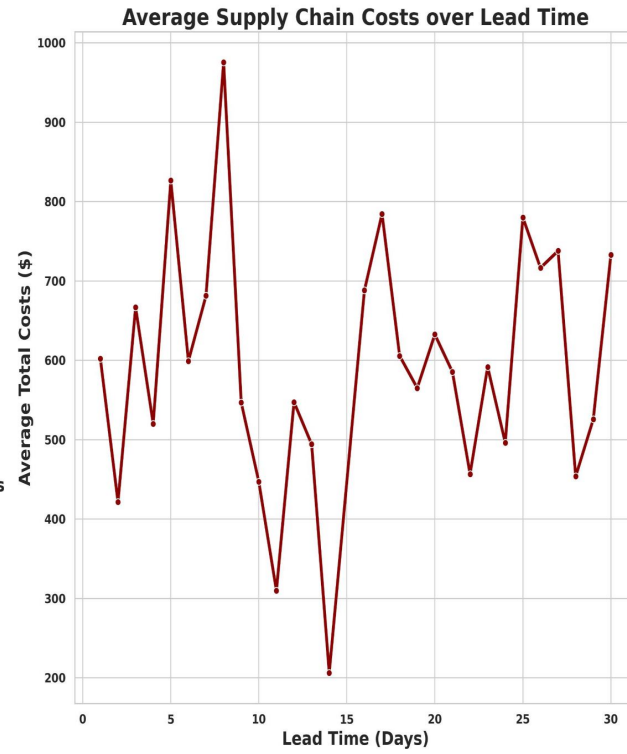


*Insights from Order Fulfillment Sheet :-

- Cosmetics Crisis: With an on-time delivery rate of only 23.1%, the Cosmetics category is the primary driver of customer dissatisfaction and requires an immediate audit.
- Haircare as a Benchmark: Haircare is the top performer (52.9% on-time); its fulfillment workflow should be used as a model for other categories.
- Systemic Inefficiency: No product category reaches a 55% on-time rate, pointing to a fundamental flaw in the overall fulfillment infrastructure.
- Fulfillment Congestion: The “Pending” status spikes dramatically at day 18 and day 28, mirroring the transportation bottlenecks and delaying order completion.
- Lead Time Instability: The high frequency of “Fail” and “Pending” statuses across the 30-day cycle suggests the fulfillment process is currently reactive rather than planned.

Supply Chain Costs Sheet





*Insights from Supply Chain Costs Sheet :-

- Operational Cost Dominance: An overwhelming 90.9% of total expenses are driven by "Other Operational Costs," suggesting that the primary budget is spent on warehousing, labor, or administrative overhead rather than direct production.
- Minimal Shipping Impact: Shipping costs represent only 1.0% of the distribution. While efficiency is important, reducing shipping rates further will have negligible impact on the overall bottom line compared to operational improvements.
- Cost Volatility vs. Lead Time: The average total costs are highly erratic, peaking sharply near day 8 (almost $1,000) and crashing at day 14 (nearly $200). This indicates inconsistent spending patterns that aren't stabilized by time.
- The "Inexpensive Window": Lead times between 11 and 15 days consistently show the lowest average costs. Investigating the processes used during this specific window could help standardize more cost-effective operations.
- Manufacturing Efficiency: At only 8.1% of the cost distribution, manufacturing is relatively lean. This suggests the business may be a distributor or assembler where the complexity (and cost) lies in the logistics and "hidden" operational tasks rather than raw creation.

*Project Conclusion :-

- The dashboard reveals that operational costs make up the largest portion of total supply chain expenses.
- Supplier performance analysis helps identify suppliers with longer delivery times and higher defect rates.
- Transportation efficiency insights show how different carriers impact transit time and delivery success rates.
- Inventory analysis highlights stock distribution across warehouses and product categories.
- Overall, the dashboard provides valuable insights to optimize supply chain operations, improve delivery reliability, and reduce defects.

Supply Chain Management Dashboard

Total Inventory Value	On-Time Delivery Rate	Average Supplier Delivery Time	Total Supply Chain Costs	Average Defect Rate
$5,000,000	95%	3 days	$2,500,000	2.28%

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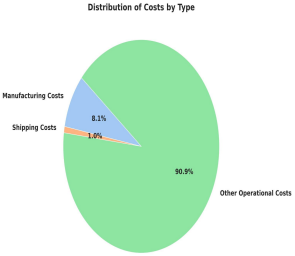

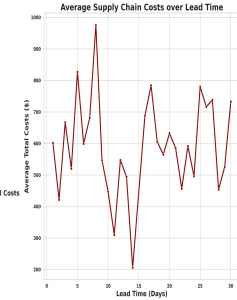

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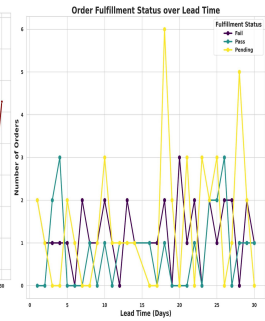

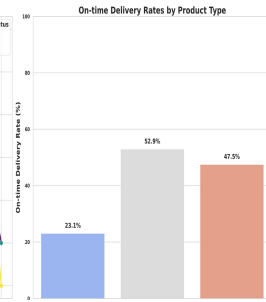

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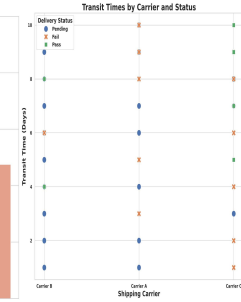

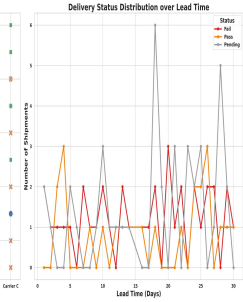

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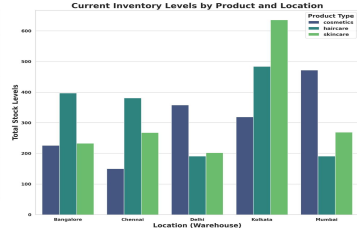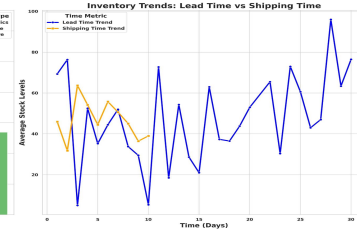
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